

Final Year Thesis/Project Report Template

by

24101661

Mohammad Asif Al Mahfuz

24241101

Rashed Satter

24241083

Sayma Shahid

22201080

Joyeta Jaglul

A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
Month Year.

© 2024. Brac University
All rights reserved.

Declaration

It is hereby declared that

1. The thesis submitted is my/our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

Student's Full Name & Signature:

Mohammad Asif Al Mahfuz

24101661

Rashed Satter

24241101

Joyeta Jaglul

22201080

Sayma Shahid

24241083

Approval

The thesis/project titled “Usable Authentication: Behavior-Based Continuous Verification” submitted by

1. Mohammad Asif Al Mahfuz (24101661)
2. Rashed Satter (24241101)
3. Sayma Shahid (24241083)
4. Joyeta Jaglul(22201080)

of [Fall-2025], [2026] has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science in [Spring-2026] Year.

Examining Committee:

Supervisor:
(Md Sabbir Ahmed)

Md Sabbir Ahmed

Senior Lecturer
Computer Science and Engineering
BRAC University

Co-Supervisor:
(Member)

Name of Co-Supervisor

Designation
Department
Institution

Thesis Coordinator:
(Member)

Name of Thesis Coordinator

Designation
Department
BRAC University

Head of Department:
(Chair)

Name of Head of Department

Designation
Department of Computer Science and Engineering
BRAC University

Abstract

In traditional authentication mechanisms, static passwords are usually used, but they compromise security or usability generally. The research study addresses behavior-based continuous authentication systems, which use user identity by using users' typing patterns, mouse dynamics, and interaction behavior. Using passive behavioral biometrics, such devices provide verification of identity assurance throughout the duration of an active session by following the Zero Trust Architecture (ZTA) philosophy of "never trust, always verify" by showing the behavioral biometrics may generate a seamless yet safe computing experience. Machine learning models are used to detect the patterns in real time to detect unauthorized user access. Experiments were performed through typical user activities, which shows that the risk of account compromise is being reduced while maintaining user convenience. This work bridges the gap between security and usability, and the potential of creating secure computing experiences for a user is discussed. Additionally, sensitive personal data is protected from unauthorized transmission. In the end, this research offers an effective and sustainable framework that paves the way for real-world implementation in individual devices and secure metaverse environments.

Keywords: Keyboard dynamics, Flight Time, SIURUA, GAIT, ZTA, FAR, FRR, Catastrophic Forgetting, ERR, Random Forest.

Table of Contents

Declaration	i
Approval	ii
Abstract	iv
Table of Contents	v
1 Introduction	1
1.1 Background	1
1.2 Rational of the Study or Motivation	1
1.3 Problem Statement	2
1.4 Objective	2
1.5 Methodology in Brief	3
1.6 Scopes and Challenges	3
2 Literature Review	5
2.1 Preliminaries	5
2.2 Review of Existing Research	5
2.3 Summary of Key Findings	9
3 Conclusion	10
References	11

Chapter 1

Introduction

1.1 Background

Authentication is the foundational process of establishing the integrity of an entity's identity before granting access to critical digital resources and information [4]. Generally, this has been gotten by passwords, which are knowledge-based, and by hardware tokens, which is possession based factor [6]. Though this method works correct in the initial stage, the whole process remains same for the authenticated users [4]. Rather than relying on a single entry point verification, Continuous Authentication (CA) or Continuous User Authentication (CUA) has come up as a dynamic solution which actually verifies identity throughout the whole session duration. This technology primarily uses behavioral biometrics such as mouse dynamics, keyboard dynamics and interaction patterns, to track unique human interaction signatures in the background with a passive way. CA system can denote the behavioral anomalies by harnessing machine learning and artificial intelligence; it really doesn't require explicit intervention of the user and adapts to user-specific baselines in real time. It offers a frictionless and seamless security experience, unlike physiological traits like fingerprints, behavioral biometrics, which rely on how a person performs[3].

1.2 Rational of the Study or Motivation

The motivation for this actually arises from the issues of frequent cyberattacks and the lack of inherent traditional security measures. According to the Reports, approximately 80% of hacking-type breaches that has brute force or compromised login information, stolen or misused credentials remain the most common first access vector in data breaches. The financial effect of these breaches are extreme. For example, Equifax data breach from 2017, did cost the company over USD 1.4 billion in settlements. Furthermore, moving to Zero Trust Architecture (ZTA) calls for a "never trust, always verify" philosophy, where, to access sensitive systems, its legitimacy must be continuously proved by every entity. Additionally, the interval for a device to be accessible after an initial login, even without the necessary additional verification process, is the aim of this research. In a word, to reduce exposure time and thus creating more user-friendly authentication structure. This study aims to mitigate the potential risks of credential theft and even session hijacking by automating the interpretation of multi-dimensional behavioral data.

1.3 Problem Statement

To assume identity wrongly throughout the session is immutable which can be done by “trust gap”, made by conventional one-time authentication mechanisms, which fail to verify identity post-unlock [20]. The password-based systems are very prone to being fragile [4]. The users sometimes strive to remember complex strings, so they are very prone to using easy-to-guess passwords, which can be attacked by social engineering attacks. Fingerprints and facial recognition are very reliable aspects of security, but they sometimes ask for active user interaction [4]. It can lead to spoofing that actually leads to being vulnerable to the presentation attacks. When a user tends to age, or faces stress or physical injury, a challenge arises in behavioral biometrics, which is called “model drift”. Additionally, standardized evaluation metrics and large-scale public longitudinal datasets and standardized evaluation metrics are missing that are specifically tailored for continuous authentication. Consequently, there is a call for a seamless, passive verification system that not only can detect intruders in real time but also do so without disrupting the user experience or compromising their privacy [20].

The study intends to look at the research issues that follow considering all the challenges. How can a seamless authentication system properly employ behavioral biometrics to ensure continuous security while effectively coping with the gradual change of a user over time?

1.4 Objective

The conception and evaluation of an efficient, safe, scalable as well as non-intrusive continuous authentication system that tracks user actions during engaged sessions represents the primary objective of this research. The study is particularly focused on these particular objectives:

- Constructing a novel machine learning or reinforcement learning-based model such as a Double Deep Q-Network (DDQN) which can use specific behavioral characteristics for example keystroke and mouse dynamics, for verifying the identity of the user.
- Implementing into operation an “Implicit Continuous User Authentication” (ICUA) pipeline which operates continuously in the background while being used.
- With applying multiple-modal biometrics for behavioral analysis, have been reported to dramatically reduce False Acceptance Rates (FAR) on compared with unimodal counterparts, the system robustness can be significantly boosted.
- Decreasing “exposure time” along with detecting unauthorized users just within a few instants of an abnormality.
- Maintaining confidential behavioral data captured during continuous observation by employing privacy-preserving techniques with the goal to tackle accessibility and confidentiality concerns.

1.5 Methodology in Brief

The structure verifies the identity of the user every time through executing five basic procedures: acquisition, preprocessing, extraction, categorization, and evaluation.

1.Acquisition: This initiates from gathering data from everyday devices with sensors including keyboards, mice, and touch screens. It does not require users to perform any actions out of the ordinary, instead it simply tracks the actions that they perform regularly like typing or moving the mouse.

2.Preprocessing: When the data has been gathered, it undergoes cleaning . This phase assures the data remains clear by eliminating any possible noise or errors and oversights. It utilizes a method that allows modification of the time of the data to ensure every detail remains entirely in alignment.

3.Extraction: The algorithm is currently searching for significant data that can identify the person using it. The result could involve the amount of time they press down keys, how rapidly they type or how they use the mouse. These insignificant characteristics vary from person to person and assists the algorithm to detect their identities.

4.Classification: After collecting those distinctive elements of details, the algorithm determines the extent to which the individual is permitted to be there or not. This investigates patterns of behavior through complex algorithms such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, or Reinforcement Learning (RL). Systems like this can develop as well as modify over time, resulting in those more effective in identifying the difference between the genuine and bogus users. The algorithm investigates the user's current actions while comparing it with the behaviors it has acquired. According to these patterns, it determines if these actions have authorization or not.

5.Evaluation :At the last stage the system's performance is examined. It monitors whether or not it's performing by considering aspects including accuracy, the False Acceptance Rate (FAR), the False Rejection Rate (FRR) and the Equal Error Rate (EER). The system can add fresh users without starting over from scratch.

1.6 Scopes and Challenges

The establishment of scalable,adaptable and secure behavioral biometric systems for mobile and desktop devices is the main goal of this research. Nevertheless there are also plenty of significant concerns that require to be resolved:

Security vs Usability :Establishing a balance between usability and security is crucial since high False Rejection Rates (FRR) irritate users while high FAR presents serious security risks.

Privacy Concerns: Private data that is sensitive need to be collected for continuous surveillance which brings up ethical issues regarding informed permission and unapproved targeting.

Behavioral Variability: Structures are changing and prone to "model drift" as the consequence of stressful situations, age-related disease as well as external factors.

Power Consumption: Mobile or computer battery life can be substantially depleted by the continuous operation of sensors and automated processing.

Adversarial Resilience: While attackers attempt to duplicate a user's behavioral characteristic, systems should be resistant in opposition to imitation and data

manipulation.

The lack of Standards: Standardized evaluation processes and data benchmarks specifically designed for ongoing authentication are insufficient.

Chapter 2

Literature Review

2.1 Preliminaries

The term “Authenticity” refers to the process of accepting the fact that an entity is claimed to be what it is. There is an identity has a set of attributes of an entity that have properties like uniqueness, universality, and acceptability. Conventionally, authentication only occurs at the beginning of a session (static authentication), and it wrongly assumes that identity is immutable post login. Continuous User Authentication (CUA) verifies identity by using behavioral biometrics, which are basically the traits that are based on how a person performs in the following actions, throughout a session [4]. These systems correspond with Zero Trust Architecture (ZTA) standards, for example, NIST Special Publication 800-207, which needs constant verification[20]. A confusion matrix provides accuracy by which the performance is evaluated, FAR (likelihood of admitting an imposter), and FRR (likelihood of rejecting a legitimate user). The Equal Error Rate (EER) exhibits the point where FAR and FRR are equal. It serves as a primary indicator of system accuracy.[4]

2.2 Review of Existing Research

According to Dmytro et al.(2022), Sometimes user’s changed physical activity can influence authentication problems, so they proposed BehaviourID which actually uses pipeline that is triggered by the app which collects data from embedded sensors and thus preprocess with Dynamic Time Warping (DTW). It executes a module which is CNN based human activity recognition module. H-MOG (100 volunteers), BB-MAS (3.5 million keystrokes) etc behavioral data were used and the performance was tested by False Acceptance rate (FAR), False Rejection Rate (FRR) and Spoof Acceptance Rate (SAR). Their proposed behaviorID achieved 0.3% FAR, 1.3% FRR, 2.5% SAR. The author addressed the need for further evaluation in unregulated real world environments and high computational overhead for the wearable devices [5].

According to Baig et al.(2021), they realised something that it is very difficult to balance security like preventing hackers, privacy like not spying on the users and not being really annoying. They found the process of tracking by mouse and typing (behavioral biometrics) are the most efficient process as this is done in the background. But, moving these data to cloud is actually risk of privacy. So, to make

this safe and secure and private, the contributed in the fact of processing these data locally on owned device through edge computing but not sending to central server. According to Kumar et al. (2006), he introduced a multimodal passive biometric system which is designed to protect interactive login sessions using face and fingerprint verification [17]. The authors built a feedback mechanism into the operating system and addressed the potential vulnerability of computers to subversion after the initial strong login. A Hidden Markov Model (HMM) is used by the methodology in order to fuse observations across modalities and time. It calculates a "Psafe" score that actually decreases in the absence of user samples to reflect the uncertainty. A linux kernel based workstation was used for performing this experiment with the training set of 500 face images and 1,000 fingerprint images per user. By stopping sensitive system calls and when the probability estimate dropped down a certain level, the system gained its security goals. However, during the heavy tasks like kernel compilation, it has brought about a performance overhead of approximately 25% [1].

An experimental research on machine learning model reinforcement learning (RL) that worked using keystroke movements for continuous authentication stated by [10]. The problem states the model's insufficiency of static methods and lack of flexibility in applications for behavioral biometric literature. The observation of typical typing patterns can be deployed by methodology like Double Deep Q-Network (DDQN) agent for the SU-AIS BB-MAS dataset (117 users) .Using PCA for the reduction of dimensional some features had been designed such as "running features" (hold times) and "summary features" (typing speed, standard deviation). The model accuracy level from 94.7% to 100% and test accuracy between 81.06% and 93.5% is demonstrated in the results. The drop behind the test accuracy is the unpredictable typing behavior of the uses that requires constant distillation to fix and identify errors.

Oladip et al(2023) has proposed a web based authentication Conventional Neural Network (CNN) . The study indicates the challenges of 2FA(two factor verification), how it is stolen credentials. In this study , data has been collected from 5 participants and 10 trails of a password. The model has 100 percent Accuracy and False Acceptance Rate(FAR) and False Reject Rate(FRR) and Equal Error Rate(EER) is zero for this paper. [9]It operates on Paillier Cryptosystem which allows users to use key encryption data without hampering the user's privacy. The trained datasets have 800 features from users' trails . This model includes Error Handling features for removing the keystrokes based on user's behavior.

A continuous learning scheme for behavioral authentication was proposed in a methodological paper, ContAuth by Chauhan et al.(2020) [3].The authors aim for two lackings, one is the depreciation in performance over time for the changes in behavior and while adding new users the model needs to be retrained from the scratch is another one. The study estimated procedures inclusive breathing, gait, and EMG using open-source longitudinal datasets. The pipeline joined deep learning (LSTM) for feature derivation with online learning (SGD) to accomplish "on-the-fly" updates. For reducing "catastrophic forgetting" the methodology engaged the iCaRL class-incremental learning method. Further research demonstrated that ContAuth m accomplished a True Positive Rate (TPR) more than 85% for breathing and gait across sessions, the performance of the merged models dropped by 45%. The system was effectively set up on an Nvidia Jetson TX2, justifying that data incremental

learning should be accomplished on a device covered by 300 seconds [3].

Saraf, G. (2026), has proposed a strong CUA framework for smartphones using multimodal behavioral biometrics to bridge the post-unlock security gap[20]. In there study, they collected synchronously touch, sliding, and inertial sensor telemetry from 15 people for creating a dataset of 35,519 samples. The study used as a methodology featuring heterogeneous deep learning architecture, which has included CNNs for sensor data, LSTMs for curvilinear sliding, and Gated Recurrent Units (GRUs) for discrete touch data. The researchers used a Grid Search Optimization (GSO) technique to identify global optima for decision thresholds to address class imbalance in Zero Trust environments. However, The optimized ensemble achieved a peak accuracy of 99.23%, and the sensor kinematics was determined as the main biometric signature. Although, it was also noted that validation needs larger and more diverse datasets [20].

Wang et al. (2022) highlighted the removal in user authentication accuracy in "hybrid scenes", which has a combination of real life tasks like typing, gaming, and browsing [7]. The problem includes the user operation changes through different scenes that creates more difficulty to simulate the patterns. However, both of the authors proposed SIURUA. This method brings out "scene-irrelated" features (which has low scenic correlation) and attached them with "user-related" features (high user correlation). Also, they performed four one hour tasks on a laptop from 41 students by which they collected the data. In the methodology for feature selection they used adjusted mutual information (AMI) and for kernel-level fusion they applied EasyMKL . Finally from the result we can see that SIURUA stands 84.0% of peak average accuracy with a 300-second time window, quitting traditional algorithms like Naive Bayes and Decision Trees. The study they showed did not have verified effects in unknown environments, but only effects for four unknown scenes [7].

Wang et al.(2024) worked on a paper to define the system security based on "Behavioral Appropriateness" to ensure the right performance on the right time. The paper demonstrates the model using the user's identity by confirming unique gait , ensuring conformity to follow system rules , and benignity for long term systemic risks [12] .The papers give us the internet penetration rate in China is 76.4 percent that urgently needs to be secure and not to rely on manual user input.The replacement of 0 percent detection rate in traditional one shot login system to with "Benignity" . The "Frictionless" CA to security in the background and change the traditional MFA for making the user's experience better. The frictionless goal is to check 100 percent security in the background. [12]

A paper written by Janani et al.(2020) proposed an idea to solve the "Data Scarcity" problem . This work introduced a Hybrid GAN VAE model for checking the behavioral data and it used Differential Privacy ($\epsilon = 1.0$) to ensure user's privacy [16]. This proposal has 96.04 percent accuracy by using the Hybrid CNN LSTM model . The reducing time of the user's spend on training models with synthetic data by 75 percent. The keystroke Dynamics are at the peak of 99.12 percent. The Enrollment Threshold is defined by using synthetic Data (GAN VAE) . This work is proving that the user's identity and personal data will not be effected.[16]

Abuhmad et al.(2020) worked on how to keep a mobile phone secure continuously by monitoring user's interactions on it. An analysis of 140 users and collecting data on Motion Based, Touch Gestures, Keystrokes Dynamics , Voice and Multimodal information.[2] This model has achieved an Equal Error Rate between 2 to 10 per-

cent. Here MultimodalSystem has more accuracy than single sensor systems over 97 percent Accuracy.[2] This model is heavily battery consuming and not able to detect “environmental noise” including users moving and drops .

A research by Ayeshwarya et al.(2024) based on the critical evaluation of “Continuous Authentication Systems”(CAS) to focus on the accuracy [11] . This is supervised learning and score level fusion combined with different biometric scores for systems classification. The paper has 122 studies of physical and behavioral biometric data on categories .Most of the studies reply on the False Rejection Rate(FRR) and Equal Error Rate (ERR).[11]

Liebers et al.(2021) researched on the modern problem like logging into Visual Reality (VR) handsets.[17] It doesn't support the typical behavioural biometric since it doesn't have physical keyboards.The study was based on 16 people while they were using simple things on VR such as pressing virtual buttons or movement of sliders . The machine learning model used here is Random Forest . The research found that every one has different hand movements and the system could identify correct users 95 percent of the time[17] . The future metaverse may not need any passwords but own gestures for identity.

Meyers et al. (2025) explained the cybersecurity as a traditional zero trust architecture, because here the entity is not that trusted. her research indicates the Ai model as the device, which analyze the contextual, behavioural and other biometrics informations to obtain the actual identity of the user [18]. In the model the author experiments and finds that the Ai driven authentications brings accuracy to 97% with a very low latency which is under 150 msec. This Ai driven authentication successfully lowers the unauthorized user's attack to 80% of that of the traditional multifactor authentication uses. Here also we can see the use of explainable Ai to ensure the trust decision of the Ai model. But after all this we can see a vulnerability in the AML in this journal. By which the attackers can use generative AI to copy the behavioural status, gesture and other necessary robustness [18].

The issue of "rigid" system was resolved by Subash et al.(2025). The vast majority of the systems follow a set of rules. for example, they may trust 50 percentage of walking. but what if he is not typing? The system might think that you may left. These writers created a context- aware 'smart system'. Which depends on how you move when you walk and how you're typing when you're seated. Their testing of this mixture produce an incredible accuracy of 98.25 percentage. This showed the need of adaptability in a safety framework [19]

Hussain et al.(2025) mainly depicts the human behavioral biometrics in his article, by using keystroke dynamics, mouse motion and movements, fingerprint, touchscreen sensor, gait pattern etc. Those steps are applied to continuously detect the attackers through the session of the overall user experience. For this, the model needed many informations, data collection, feature extraction etc. which were maintained by some algorithms, such as (SVM) and the neural networks as well. This artical hase some restriction to authentications. Like, when user get stressed, fatigue, or fever, serious injuries, mental problem, the authentications proccess may hamper. For this they didnt took any steps [15].

Bennett and Carter bring out the sensor fusion concept as a comprehensive review of the AI mechanism. Their article depicts that how a holistic user profile can be created using gyroscope, GPS, accelerometers etc. Here, author gave importance to the Risk Adaptive Authentication, as the AI detects the difference from the exact

behavioural biometrics. There is also a concern that they missed in their article is the battery health of its battery and hardware system [13].

2.3 Summary of Key Findings

From all of those journals we get to know that, it shows the needed change from the system of authentication of episodic or point of entry to continuous behavioural validation platform. Also we can see here they initially adopted deep learning architectures, like CNNs and LSTMs, which are expert in solving complex high defined sensors from the data, like touch gesture and walking gestures. Moreover, this Multi-modal fusion is mostly known as a powerful strategy, where we can combine multiple behavioural signals (e.g., keystroke + mouse + gait). This effectively deducts invalid acceptance rates and enhances protections to mimicry attacks. Meanwhile the research also depicts the importance of zero trust environment, which shows us that the trust must be continuously proven by the user behavioural activities. However, the controlled research was 95% correct in most of the cases, but in real life implementations there remain some challenges. User behavior is not static; symptoms like fatigue, stress, and aging cause "model drift," which turns into low performance over a long period of time. Though it creates difficulty to add new users for the "Catastrophic Forgetting" issue in neural networks, Continuous Authentication shows hopeful results to incremental learning frameworks. Privacy is also important, because this kind of system can collect the information of the users. That's why we need safe models like homomorphic encryption and federated learning. Lastly, there is not any single valid dataset or examination system. That's why it gets complicated to compare with different types of research.

Chapter 3

Conclusion

In conclusion, we can say that the concept of point-of-entry authentication like passwords and PINs, is not safe for modern life styles because it struggles to identify the actual identity of the user and can be the equal for the whole session. As Continuous Authentication passively monitors behavioral biometrics it effectively closes the trust gap. Like, the motion of users typing, the movement of the mouse and how the users swipe the screen. It verifies the user's identity on the fly. Those journals depict that some sort of AI models, such as deep learning model, Reinforcement Learning model(which is reward giving) can accurately identify the exact user. The accuracy often crosses over 95%, when it learns these unique behavioral fingerprints. Also, here in research multimodal is used to combine multiple behavioral fusions, which helps to lower the risk of an attacker from hijacking information or any misdeed, because it is quite impossible for an attacker to copy those multiple behaviors together. Though the security makes security easier for the users, users also have to think about "Model Drift", that happens when a user's health gets worse and behaviour changes because of aging, stress, tension etc. So, to preserve those important and sensitive information of users, there we can see the use of privacy tools, such as homomorphic encryption and on-device processing, which make sure that the personal information of the users will be safe and never be stolen at any condition. This study demonstrates that behavior-based continuous verification is an essential step toward a safe and user friendly "Zero Trust" future by lowering the "exposure time", the duration of time the device gets exposed after being unlocked and so that it can be close to zero.

Bibliography

- [1] S. Kumar, T. Sim, R. Janakiraman, and S. Zhang, “Using continuous biometric verification to protect interactive login sessions,” in *21st Annual Computer Security Applications Conference (ACSAC)*, 2006. DOI: 10.1109/csac.2005.61.
- [2] M. Abuhamad, A. Abusnaina, D. Nyang, and D. Mohaisen, “Sensor-based continuous authentication of smartphones’ users using behavioral biometrics: A contemporary survey,” *arXiv.org*, 2020. [Online]. Available: <https://arxiv.org/abs/2001.08578>.
- [3] J. Chauhan, Y. D. Kwon, P. Hui, and C. Mascolo, “Contauth: Continual learning framework for behavioral-based user authentication,” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 4, no. 4, p. 122, 2020. DOI: 10.1145/3432203.
- [4] A. F. Baig and S. Eskeland, “Security, privacy, and usability in continuous authentication: A survey,” *Sensors*, vol. 21, no. 17, p. 5967, 2021. DOI: 10.3390/s21175967.
- [5] D. Progonov, V. Cherniakova, P. Kolesnichenko, and A. Oliynyk, “Behavior-based user authentication on mobile devices in various usage contexts,” *EURASIP Journal on Information Security*, vol. 2022, no. 1, 2022. DOI: 10.1186/s13635-022-00132-x.
- [6] I. Stylios, S. Kokolakis, O. Thanou, and S. Chatzis, “Key factors driving the adoption of behavioral biometrics and continuous authentication technology: An empirical research,” *Information and Computer Security*, vol. 30, no. 4, pp. 562–582, 2022. DOI: 10.1108/ics-08-2021-0124.
- [7] X. Wang, Y. Shi, K. Zheng, Y. Zhang, W. Hong, and S. Cao, “User authentication method based on keystroke dynamics and mouse dynamics with scene-irrelated features in hybrid scenes,” *Sensors*, vol. 22, no. 17, p. 6627, 2022. DOI: 10.3390/s22176627.
- [8] A. F. Baig, S. Eskeland, and B. Yang, “Privacy-preserving continuous authentication using behavioral biometrics,” *International Journal of Information Security*, vol. 22, no. 6, pp. 1833–1847, 2023. DOI: 10.1007/s10207-023-00721-y.
- [9] I. D. Oladipo, M. Nicho, J. B. Awotunde, J. O. Buari, M. Abdulraheem, and T. Gaber, “A secure and scalable behavioral dynamics authentication model,” *International Journal of Advanced Computer Science and Applications*, vol. 14, no. 8, pp. 21–22, 2023. [Online]. Available: <https://www.ijacsa.thesai.org>.

- [10] P. Bansal and A. Ouda, "Continuous authentication in the digital age: An analysis of reinforcement learning and behavioral biometrics," *Computers*, vol. 13, no. 4, p. 103, 2024. DOI: 10.3390/computers13040103.
- [11] A. Sundar and K. J. Singh, "A comprehensive review on secure biometric-based continuous authentication and user profiling," *IEEE Access*, vol. 12, pp. 82 996–83 021, 2024. DOI: 10.1109/access.2024.3411783.
- [12] C. Wang, H. Tang, H. Zhu, J. Zheng, and C. Jiang, "Behavioral authentication for security and safety," *Security and Safety*, vol. 3, 2024. DOI: 10.1051/sands/2024003.
- [13] D. Carter, S. Martin, and K. Brooks, "Ai-based continuous authentication systems," *ResearchGate*, 2025. [Online]. Available: <https://www.researchgate.net/publication/397880741>.
- [14] C. Gilbert and M. A. Gilbert, "Continuous user authentication on mobile devices," *International Research Journal of Advanced Engineering and Science*, vol. 10, no. 1, pp. 158–173, 2025. [Online]. Available: <https://irjaes.com/wp-content/uploads/2025/03/IRJAES-V10N1P355Y25.pdf>.
- [15] S. Hussain, "Behavioral biometrics and continuous authentication in cybersecurity systems," *ResearchGate*, Jun. 2025. DOI: 10.13140/RG.2.2.35971.41763.
- [16] K. Janani, "The human-machine identity blur: A unified framework for cybersecurity risk management in 2025," *arXiv.org*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.18255>.
- [17] G. Kang, J. Park, and Y.-G. Kim, "Continuous behavioral biometric authentication for secure metaverse workspaces in digital environments," *Systems*, vol. 13, no. 7, p. 588, 2025. DOI: 10.3390/systems13070588.
- [18] J. R. Meyers, A. D. Thompson, M. L. Greene, and C. James, "Ai-based continuous authentication in zero trust environments," *ResearchGate*, Jun. 2025. [Online]. Available: <https://www.researchgate.net/publication/396744658>.
- [19] A. Subash, I. Song, I. Lee, and K. Lee, "Adaptability of current keystroke and mouse behavioral biometric systems: A survey," *Computers Security*, 2025. DOI: 10.1016/j.cose.2025.104731.
- [20] G. Sarraf, "Behavioral analytics for continuous insider threat detection in zero-trust architectures," *arXiv.org*, 2026. [Online]. Available: <https://arxiv.org/abs/2601.06708>.